

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.3: Vectors I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Vectors I
Driving Question	DRIVING QUESTION: Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Scalars vs. Vectors — Launching the Ferry Problem	A passenger ferry departed Kisumu heading due east toward Mbita but arrived far south of the intended destination because a southward lake current was also acting on the vessel at the same time.	Quantities that have only magnitude are called scalars. Quantities that have both magnitude and direction are called vectors. Direction is not optional information — it is mathematically essential for describing motion, force, and displacement.	The ferry missed Mbita because the engine thrust (a vector pointing east) and the lake current (a vector pointing south) both acted simultaneously. Since direction matters, the two pushes combined to produce a resultant journey that was neither purely east nor purely south — but diagonally south-east, landing the ferry at the wrong point on the shore.
Lesson 2 Representing Vectors — Column Vectors and Position Vectors	Learners observed that any directed arrow on a grid can be fully described by counting its horizontal steps (right positive, left negative) and vertical steps (up positive, down negative), and recording these as a column vector. They observed that the ferry's engine thrust produces a column vector of $(20, 0)$ — eastward only — while the lake current produces $(0, -8)$ — southward only — and that these two separate column vectors sit inside the same ferry diagram.	Learners learned that a column vector is a precise mathematical notation that captures both the magnitude and direction of a vector using two signed components. They learned that a position vector describes the location of a point relative to an origin, written in column vector form. They connected these ideas to the ferry scenario by writing the position vector of Mbita as $(60, 0)$ and recognising that the ferry's actual landing point has a non-zero negative bottom entry — a direct mathematical consequence of the southward current.	Learners explained why the captain's intended path can be written as $(60, 0)$ — with no vertical component — while the actual journey must include a negative vertical component due to the current. They articulated that column vector notation is the mathematical language needed to describe the ferry's drift precisely, and that combining the engine vector and the current vector into one resultant vector is the next mathematical step — which they identified as vector addition.
Lesson 3	Learners observed two simultaneous vector	Learners learned that two vectors can be	Learners explained that the Kisumu ferry

Vector Addition — Triangle and Parallelogram Rules	diagrams of the Kisumu–Mbita ferry — an eastward engine thrust (20 km/h) and a southward current (8 km/h) — and constructed both the triangle-law and parallelogram-law diagrams on grid paper, measuring the resultant arrow's length and direction with rulers and protractors.	combined into a single resultant vector using the triangle law (tip-to-tail) or the parallelogram law (diagonal of adjacent sides), that both methods yield the same resultant, that the magnitude can be calculated using the Pythagorean theorem ($ R = \sqrt{20^2 + 8^2} \approx 21.5$ km/h), and that the direction is found using $\arctan$ ($\theta = \arctan(8/20) \approx 21.8^\circ$ south of east).	missed Mbita because the southward current vector combined with the eastward engine vector to produce a resultant displacement directed approximately 21.8° south of east — the ferry did not travel due east but along the resultant path, landing south of the intended destination by a calculable amount.
Lesson 4 Subtraction of Vectors and the Zero Vector — Charting the Correction Course Back to Mbita	The ferry landed at point L, which is 24 km south of Mbita (point M), after the engine vector and current vector were combined. A dashed arrow on the diagram connected L to M and had unknown components. A Nakuru farmer also overshot a maize store when pulling a rope cart, landing at the wrong point.	The negative of a vector reverses its direction while keeping the same magnitude. The zero vector results from adding a vector to its negative and represents no net displacement — a round trip. Vector subtraction $a - b$ is the same as $a + (-b)$ and is carried out component by component in column vector notation. The correction vector from L to M equals the column vector (0 above 24), found by subtracting the position vector of L from the position vector of M.	The captain can calculate exactly what correction vector is needed to guide the ferry from its wrong landing point to Mbita by subtracting position vectors. The mathematics of vector subtraction gives a precise, reproducible tool for correcting any navigation error caused by drift — whether from a lake current, ocean tide, or wind — and ensures the ferry can always reach the right destination.
Lesson 5 Scalar Multiplication of Vectors and Parallel Vectors	Learners drew scaled vectors geometrically (v , $2v$, $-v$) using ruler and scale, calculated scalar multiples of column vectors ($2p$, $-3q$, $\frac{1}{2}r$), and tested four vector pairs for parallelism by finding scalar k — including identifying a non-parallel pair (e and f) where component ratios were unequal.	Multiplying a vector by a positive scalar scales its magnitude while preserving its direction; multiplying by a negative scalar reverses the direction; two vectors are parallel if and only if one is a scalar multiple of the other (with the sign of k indicating same or opposite sense); these operations model real changes in the ferry scenario such as a stronger current or increased engine thrust.	Learners explained why the Kisumu ferry would drift even further south if the lake current were scaled up (e.g., $1.5c$ or $2c$), calculated the resulting drift distance using scalar multiplication, identified the engine's original and increased thrust vectors as parallel (same direction, different magnitude), and recorded a Captain's Advisory quantifying the increased south-drift for a doubled current.
Lesson 6 Magnitude of a Vector — How Far Did the Ferry Really Travel?	The ferry departed Kisumu heading due east at 20 km/h while a southward Lake Victoria current of 8 km/h simultaneously pushed it off course. The resultant journey formed a right-angled triangle with legs of 20 km/h (east) and 8 km/h (south).	The magnitude of a resultant vector formed from two perpendicular components is calculated using the Pythagorean theorem: $ v = \sqrt{x^2 + y^2}$. For the ferry, $ v = \sqrt{20^2 + 8^2} = \sqrt{464} \approx 21.5$ km/h, meaning over 3 hours the ferry actually travelled approximately 64.6 km — not the 60 km straight-line engine distance.	Because the engine thrust (east) and the current (south) are perpendicular directed quantities, they combine as the two legs of a right-angled triangle. The actual path of the ferry is the hypotenuse of that triangle. Pythagoras gives us the exact length of the hypotenuse — and therefore the true distance travelled. This is why simply adding 20 and 8 is mathematically wrong: direction forces us to use squares and square roots, not plain addition.
Lesson 7 Collinear Points and	Students observed that when three ports — Kisumu K(0,0), Kendu Bay B(4,2), and	Students learned that three points are collinear if and only if the vectors	Students explained that the Kisumu ferry captain can use the scalar multiple

Vector Paths	Mbita M(8,4) — are plotted on a coordinate grid, the vectors KB and KM appear to lie along the same straight line. They also observed that for a second set of ports P(1,3), Q(3,7), R(5,10), the visual alignment is less obvious, requiring algebraic confirmation.	connecting them from a common point are scalar multiples of each other — meaning one vector is a stretched or shrunk version of the other in exactly the same direction. They also learned that a multi-stop ferry journey can be expressed as a vector chain (the sum of sequential displacement vectors), and that this chain simplifies to a single resultant displacement vector from start to finish.	collinearity test to verify before sailing whether a planned multi-stop route is truly a straight-line path, avoiding unnecessary course corrections. They also explained that by expressing the full journey as a vector chain, the captain can calculate the exact total displacement from port of departure to final destination — without needing to measure each leg separately — giving precise, mathematically guaranteed navigational information.
Lesson 8 Translation Vectors and Unit Synthesis — Solving the Ferry Problem	Students observed their own Lesson 1 prediction drawings — simple, undirected east-pointing arrows — alongside the complete vector diagram of the ferry problem showing the engine vector [20, 0], the current vector [0, -8], and the resultant vector pointing south of east with magnitude ≈ 21.5 km/h at a bearing of $\approx 111.8^\circ$. They observed the DQB populated across 8 lessons, audited every question card with a traffic-light system, and witnessed all cards reach GREEN status through student-generated answers.	Students learned that: (1) two simultaneous vectors — the engine's eastward push and the current's southward pull — combine by addition into a single resultant vector, explaining exactly why the ferry missed Mbita; (2) the magnitude of the resultant (≈ 21.5 km/h, true distance ≈ 64.6 km) is found using Pythagoras' theorem on the component vectors; (3) the drift angle ($\approx 21.8^\circ$ south of east, bearing $\approx 111.8^\circ$) is found using trigonometry; (4) the corrective heading the captain needs (bearing $\approx 066.4^\circ$, angled $\approx 23.6^\circ$ north of east) is found by reverse-engineering the vector addition so the northward component of engine velocity exactly cancels the southward current; (5) vectors differ from scalars because direction is mathematically encoded and changes the physical outcome — pointing east is not the same as pointing east while accounting for the current.	Students explained that the Kisumu ferry missed Mbita because the captain treated velocity as a scalar (speed only, pointing east) without accounting for the current as a second vector simultaneously acting on the vessel. The resultant of the two vectors pointed south of east, landing the ferry kilometres below Mbita. Using the full toolkit of Vectors I — column vector representation, vector addition, Pythagoras' theorem, trigonometric direction finding, and vector decomposition — the captain can calculate the precise corrective bearing that pre-compensates for any known current or wind, ensuring the ferry always arrives at the intended destination. This is why direction is not optional in describing real-world quantities: mathematics with direction saves time, resources, and lives.

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